

STUDENT SUPPORT INSIGHTS TOOL

AI-Enabled Learner Support Analytics Prototype

Final Integrated Report | SFIA Level 3 Integrated Assessment | Programme AI Internship - Cohort 5

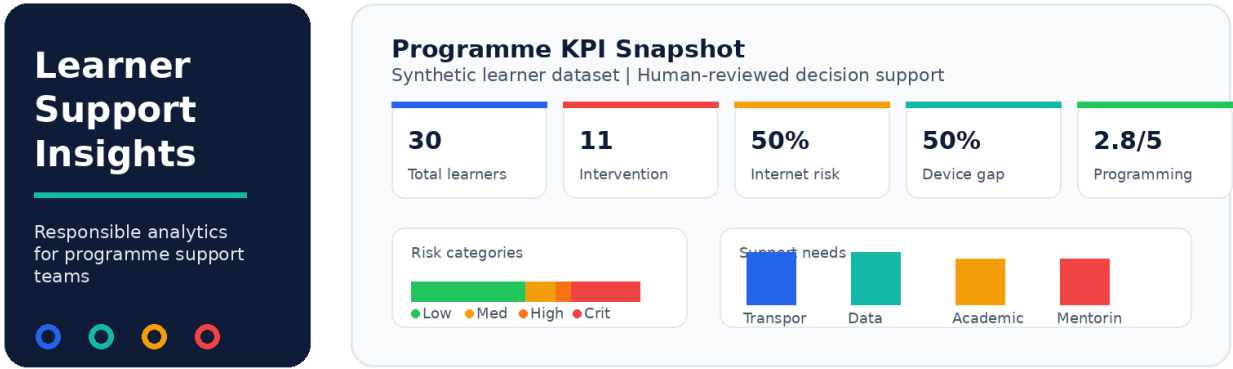


Table of Contents

1. Executive Summary	3
2. Problem Statement and Context	4
3. Stakeholder and User Needs Analysis	5
4. Current and Improved Process	6
5. Business and Value Model	8
6. Data Description and Preparation	9
7. Analytics Findings and KPI Dashboard	11
8. Prototype Description and UX/UI Design	15
9. AI, Data Ethics and Privacy Review	18
10. Methods, Tools and Development Approach	20
11. Testing, Review and Quality Evidence	22
12. Hosting and Deployment Readiness	24
13. Reflection and Future Improvements	26
14. Evidence Inventory Mapped to SFIA Skills	27
15. Final Submission Checklist	29
Appendices	30

Document Control

Item	Description
Report purpose	Integrated final report for the Student Support Insights Tool assignment.
Evidence basis	Project source code, dataset, documentation, screenshots, testing logs, bug logs, ethics review and SFIA evidence checklist.
Data status	Synthetic learner records only. No real learner personal information is used.
Decision status	Risk flags and recommendations are decision-support signals only and require human staff review.

1. Executive Summary

This report presents the Student Support Insights Tool, a working prototype designed to help programme staff capture, validate, analyse and interpret learner support survey data. The solution turns raw learner support information into operational insight through KPI cards, visual dashboards, explainable risk scoring, recommendation outputs and responsible-use controls.

The project responds to a common challenge in digital skills programmes: learner information is often collected, but it is not always used systematically to identify early support needs, access barriers or process improvement opportunities. This prototype demonstrates how a programme team can move from manual spreadsheet review to proactive, ethical and data-informed learner success management.

The prototype uses a synthetic dataset of 30 learners. The risk model identifies 11 learners who require high or critical staff review, while also highlighting connectivity risk, device access gaps and confidence patterns across digital, programming and AI familiarity skills. These outputs are not final decisions; they are structured evidence for staff review and support planning.

The final evidence pack includes source code, dataset files, data dictionary, process models, business value model, ethics and privacy review, testing evidence, bug log, screenshots, README/user guide, peer/stakeholder review evidence and SFIA mapping. Together, these artefacts show a complete integrated assessment outcome rather than disconnected mini-submissions.

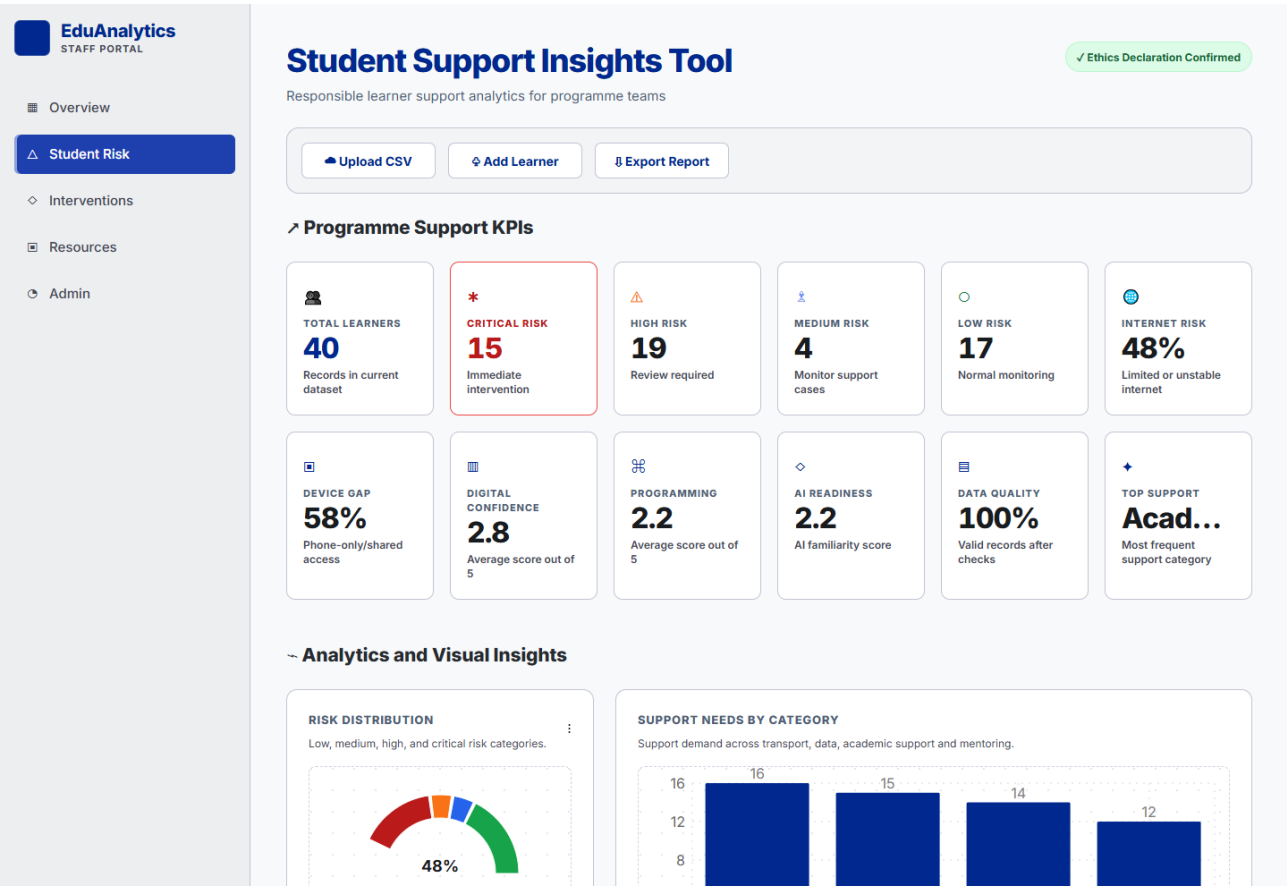


Figure 1: Dashboard overview showing the operational layout and learner support KPI story.

2. Problem Statement and Context

Digital skills programmes depend on timely learner support. Learners may face barriers such as limited data, unstable internet, phone-only learning, transport difficulty, low confidence or limited familiarity with AI tools. When this information sits in spreadsheets or disconnected notes, programme staff may not see risk patterns early enough to act.

The core project question is: How can a programme team use learner data responsibly to identify support needs, prioritise support actions and improve learner success?

Current challenge	Impact on programme delivery	How the prototype responds
Manual learner support review	Staff spend time scanning spreadsheets and may miss early warning signs.	The tool validates data and summarises risk through dashboard KPIs.
Inconsistent support prioritisation	Support decisions can vary depending on who reviews the data.	A transparent scoring model gives consistent risk categories and reason codes.
Limited reporting visibility	Leadership may not see the full pattern of learner needs.	Charts and summaries translate data into management-level insight.
Ethical risk in using learner data	Learners can be unfairly labelled if outputs are treated as final decisions.	Human review, consent wording and non-automation rules are built into the workflow.

3. Stakeholder and User Needs Analysis

The tool is designed primarily for programme staff who need to understand learner support needs quickly and fairly. It also serves leadership, facilitators, mentors and IT support by creating a clearer evidence trail from data capture to support action.

Stakeholder	Need	Design response
Learners	Fair, timely and relevant support without unnecessary exposure of personal data.	Synthetic/anonymised data approach, privacy wording and human review.
Programme staff	A simple way to identify support patterns and prioritise action.	KPI dashboard, risk table, recommendation cards and validation feedback.
Facilitators / mentors	Clear signals about who may need academic or confidence support.	Reason codes and support categories make follow-up easier.
Programme leadership	Management view of support demand, risk and resource needs.	Trends, KPI cards and exportable summaries support reporting.
IT / support administrators	A maintainable prototype with clear setup instructions.	React frontend, ASP.NET Core API, README, environment settings and build evidence.

User Personas

Persona	Primary goal	Success measure
Student Support Officer	Find learners needing urgent support and understand why.	Can filter risk review table and see reason codes.
Programme Manager	Plan support resources using reliable evidence.	Can view top needs, device gaps and internet risk.

Facilitator	Adjust teaching support based on learner readiness.	Can see confidence trends and academic support needs.
Assessor / reviewer	Verify that the assignment evidence is complete.	Can trace every SFIA requirement to evidence files.

4. Current Process and Improved Process

The current process is mostly manual and reactive. The improved process introduces structured data validation, dashboard analysis, risk flagging and staff-reviewed recommendations. This creates a more accountable support process while keeping human judgement in every learner decision.

Current As-Is Learner Support Process

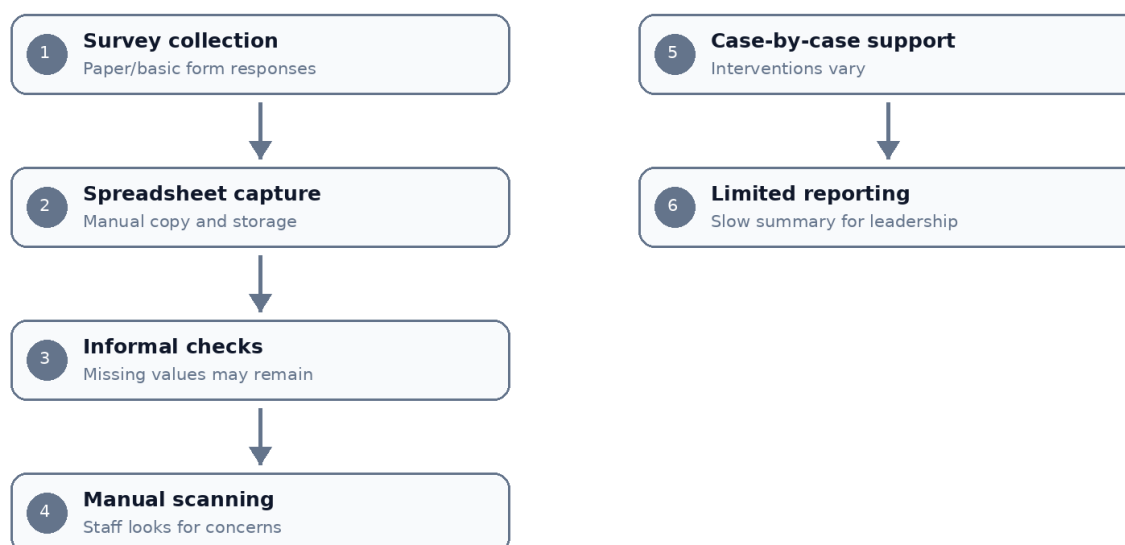


Figure 2: Current as-is process, showing manual review and weak feedback loops.

Improved To-Be Learner Support Process

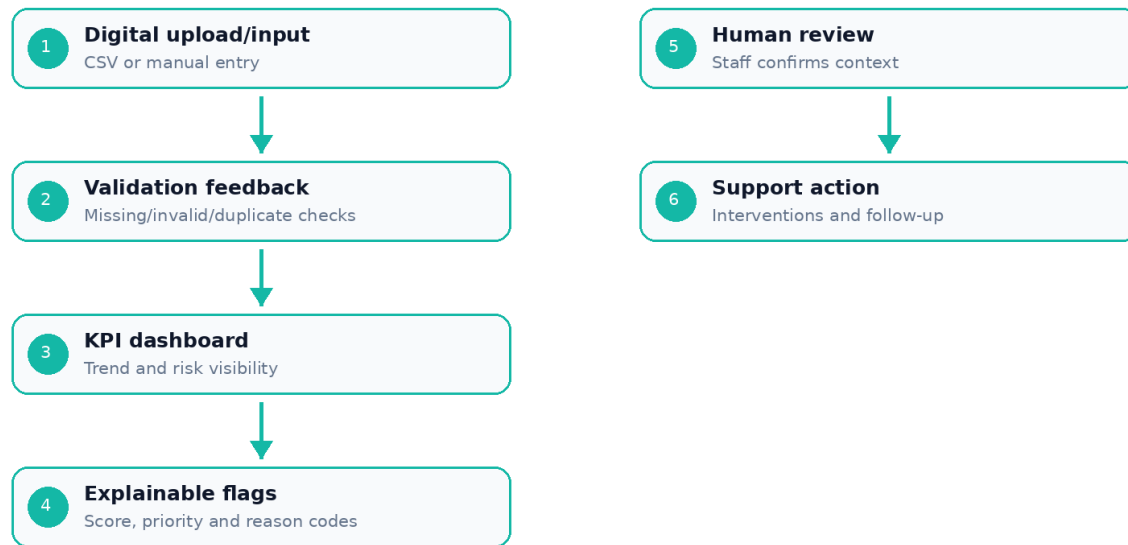


Figure 3: Improved to-be process, showing validation, analysis, risk review and follow-up.

Process Improvement Opportunities

Improvement opportunity	Operational value	Evidence in prototype
Automated validation	Improves data quality before analysis begins.	CSV upload and manual form validation.
Early risk visibility	Supports proactive intervention rather than late response.	Risk score, priority and reason codes.
Clear management reporting	Improves decision-making and resource allocation.	KPI dashboard and five visualisations.
Responsible-use workflow	Prevents automatic labelling or unfair treatment.	Ethics declaration and privacy notice.
Feedback loop	Encourages staff to review, act and monitor outcomes.	To-be process and recommendation cards.

5. Business and Value Model

The value model shows how learner survey data becomes a practical programme management asset. The tool does not only display charts; it supports resource planning, prioritised staff review, targeted intervention and continuous improvement.

Value Model: From Data to Learner Support Action

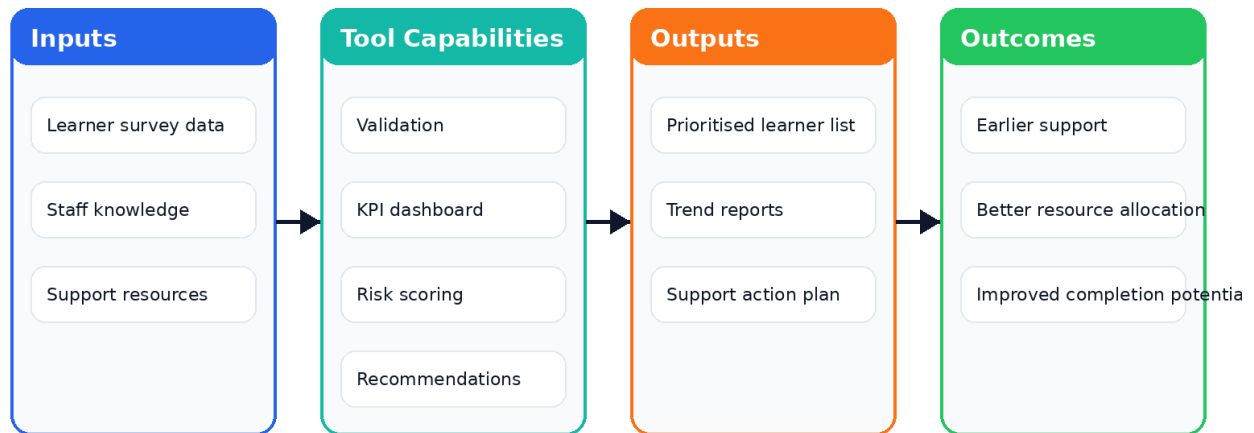


Figure 4: Value model linking data inputs to programme outcomes.

Value area	What the tool improves	Possible indicator
Efficiency	Reduces manual spreadsheet scanning.	Time taken to produce support summary.
Quality	Standardises validation and risk interpretation.	Number of invalid records detected before reporting.
Responsiveness	Highlights high and critical cases early.	Number of flagged learners reviewed weekly.
Resource planning	Shows support needs by category and access barrier.	Allocation of data, device, transport and mentoring support.
Accountability	Provides evidence for decisions and follow-up.	Review notes and exported summaries.

6. Data Description and Data Preparation

The dataset contains 30 synthetic learner records. Synthetic data was used to demonstrate the tool safely without exposing real learner personal information. Each learner record includes access conditions, confidence scores, employment status, support need, attendance risk and a short notes field.

Field	Description	Example
LearnerID	Unique learner reference.	L001
AgeBand	Age group category.	18-24
Province	Learner area/province.	Western Cape
DeviceAccess	Primary learning device.	Laptop, Shared laptop, Phone only
InternetAccess	Connectivity status.	Reliable, Unstable, Limited
DigitalConfidence	Self-rated digital confidence from 1 to 5.	4
ProgrammingConfidence	Self-rated programming confidence from 1 to 5.	3

AI Familiarity	Self-rated AI familiarity from 1 to 5.	2
Employment Status	Employment category.	Unemployed
Support Need	Primary support need.	Data, Transport, Academic support, Mentoring
Attendance Risk	Attendance risk category.	Low, Medium, High
Notes	Short context note.	Needs bus pass

Data Preparation and Validation

- Missing value checks for required learner fields.
- Controlled-value validation for age band, device access, internet access and attendance risk.
- Confidence score range validation from 1 to 5.
- Duplicate learner ID detection.
- Quote-aware CSV parsing so commas inside notes do not break import.
- Row-level validation feedback with line numbers and field names.
- Inline validation for manual learner entry.

Data Quality & Validation



OVERALL HEALTH

1 records loaded. 8 validation issue(s) currently visible.

ROW	FIELD	ISSUE	SUGGESTED FIX
2	Employment status	Employment status is required.	Correct the value and upload again.
2	Age band	Age band must be one of 18-24, 25-29, or 30-35.	Correct the value and upload again.
2	Device access	Device access must be Laptop, Shared laptop, or Phone only.	Correct the value and upload again.
2	Internet access	Internet access must be Reliable, Unstable, or Limited.	Correct the value and upload again.

Figure 5: Upload validation evidence showing row-level feedback.

7. Analytics Findings and KPI Dashboard

The dashboard exceeds the minimum visualisation requirement by providing five charts and a richer KPI layer. The analytics are designed for programme staff, not technical users, so labels, colours and helper text focus on operational meaning.



Figure 6: KPI snapshot based on the synthetic learner dataset.

KPI Summary

KPI	Result	Interpretation
Total learners	30	Synthetic learner records available for analysis.
Critical risk	9	Learners requiring immediate staff review.
High risk	2	Learners requiring urgent follow-up.
Medium risk	4	Learners needing monitoring and early support.
Low risk	15	Learners currently showing lower support risk.
High + critical review group	11	Learners requiring priority intervention planning.
Internet risk	15 learners (50%)	Limited or unstable connectivity affects half the dataset.
Device access gap	15 learners (50%)	Phone-only or shared laptop access affects half the dataset.
Average digital confidence	3.2 / 5	Moderate readiness.
Average programming confidence	2.8 / 5	Clear need for coding support.
Average AI familiarity	2.3 / 5	Lowest readiness area.
Data quality score	100%	The synthetic dataset is complete and valid for prototype analysis.

Five Data-Driven Insights

- 11 of 30 learners fall into high or critical review categories, making early support planning a priority.
- 15 learners report limited or unstable internet, which is a major access barrier.

3. 15 learners use either a phone only or a shared laptop, showing a significant device access gap.
4. Transport and data are the strongest support categories, with 8 transport cases and 8 data cases.
5. AI familiarity is the lowest average confidence area at 2.3 out of 5, followed by programming confidence at 2.8 out of 5.

Practical Recommendations

Recommendation	Rationale	Action
Connectivity and device support package	Internet risk and device gaps affect half of the dataset.	Prioritise data vouchers, offline learning packs and device-lending slots.
Targeted attendance support workflow	High and critical learners show combined attendance, access and confidence risk factors.	Assign mentor check-ins, transport follow-up and weekly review meetings.
Confidence-building academic clinics	Programming and AI familiarity are lower than general digital confidence.	Run short practical labs, peer tutoring and beginner-friendly AI/coding sessions.
Resource planning dashboard review	Support needs are concentrated enough to guide planning.	Review KPI dashboard weekly and export summary for programme meetings.

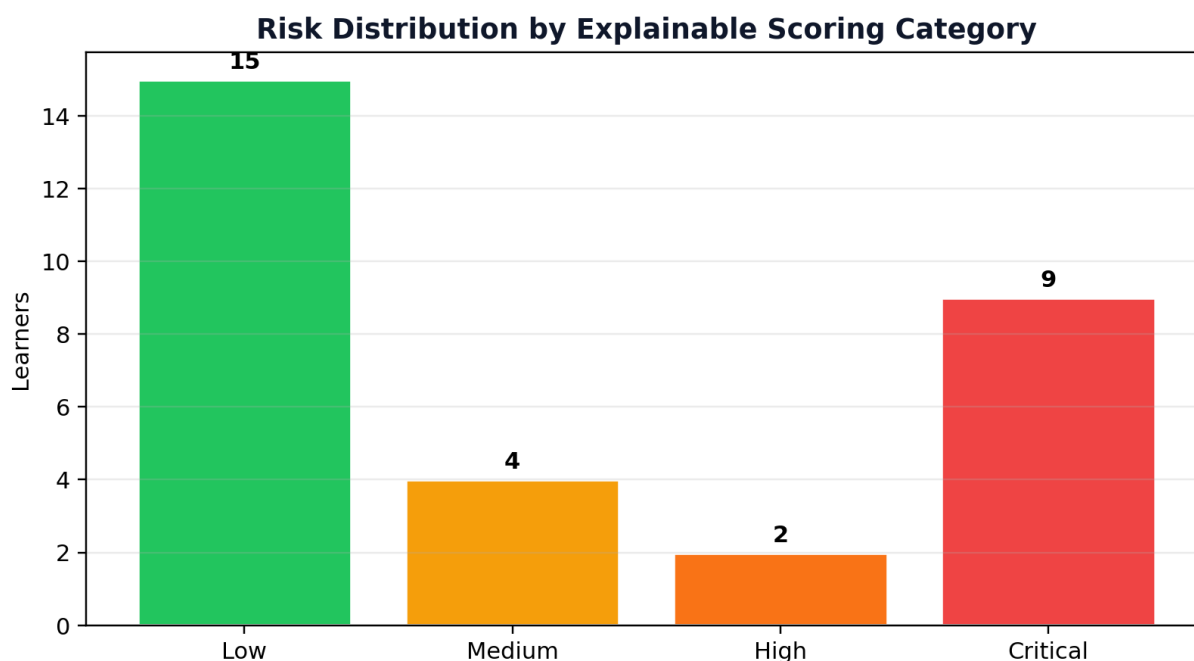


Figure 7: Risk distribution chart.

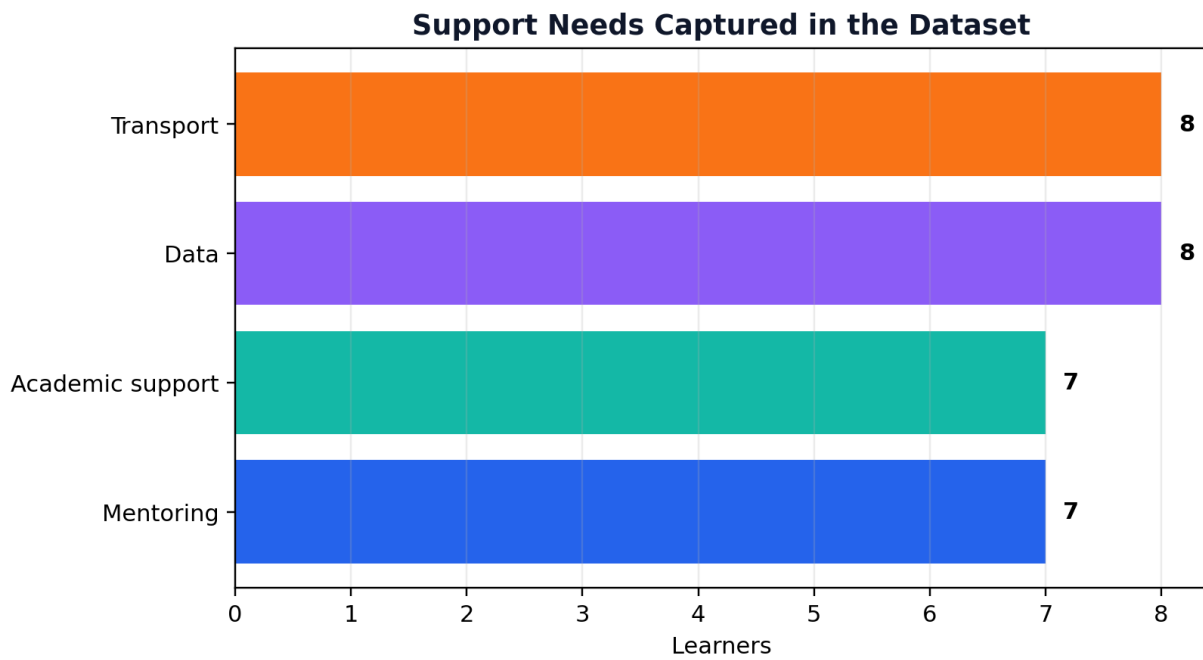


Figure 8: Support needs breakdown.

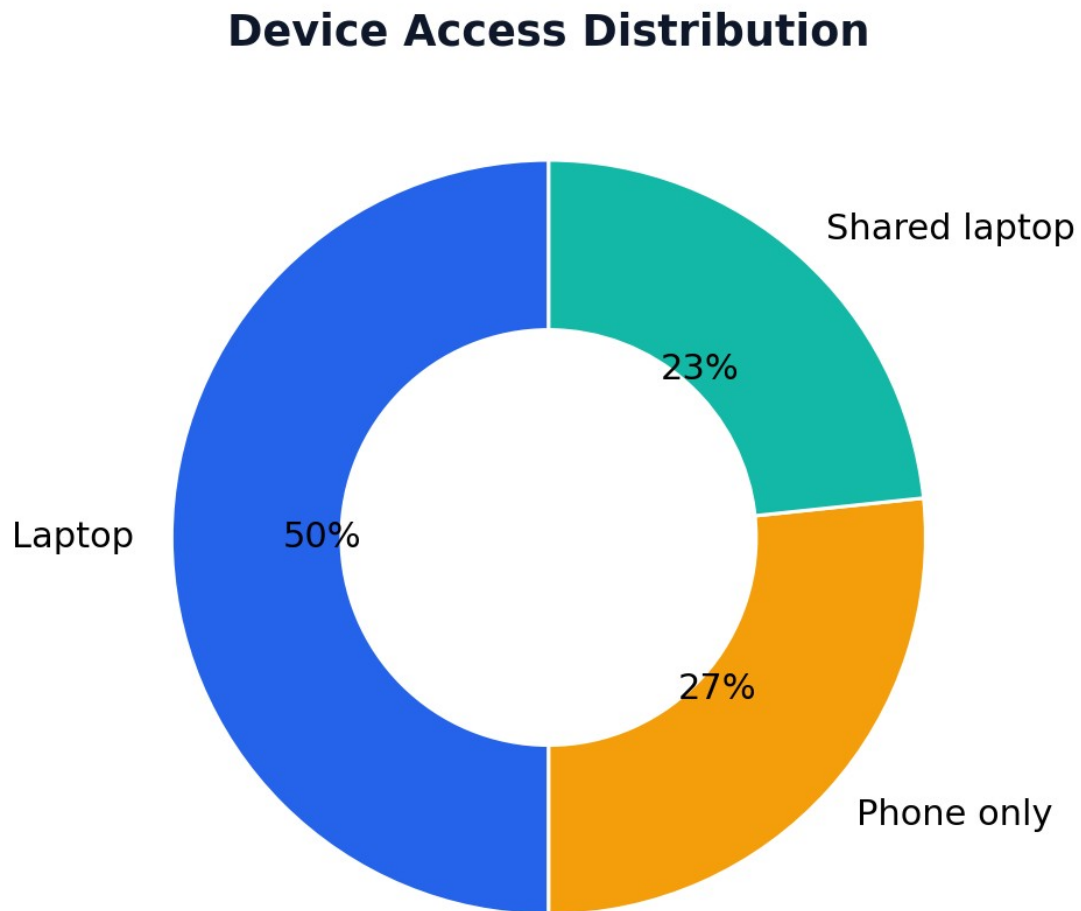


Figure 9: Device access distribution.

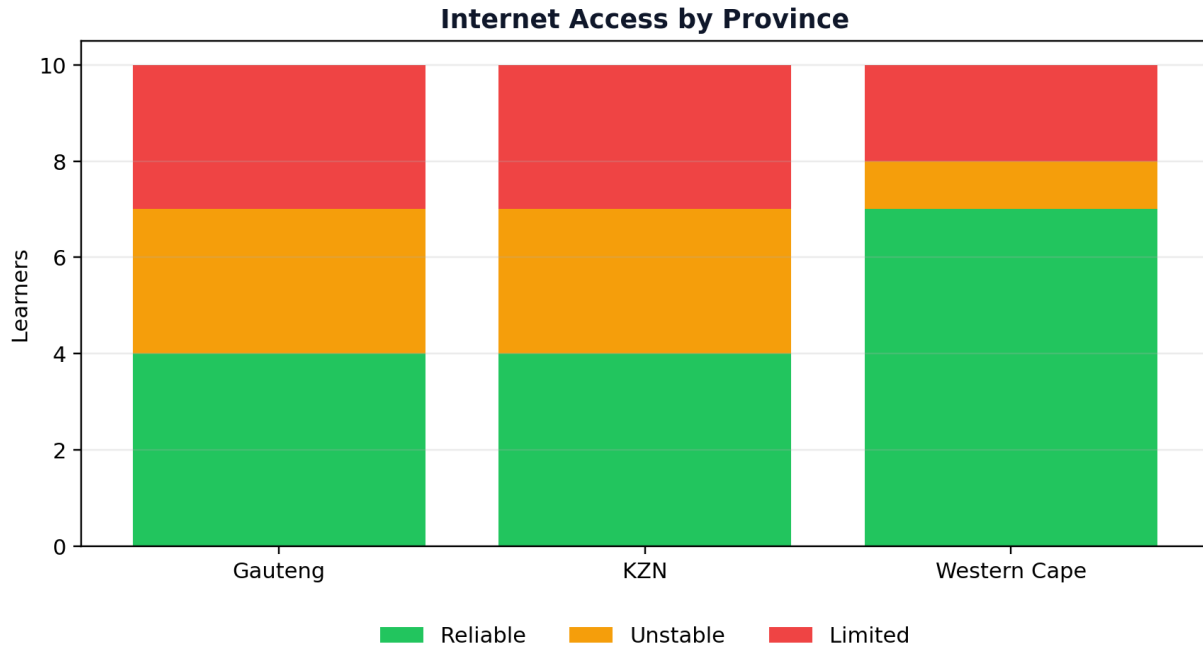


Figure 10: Internet access by province.

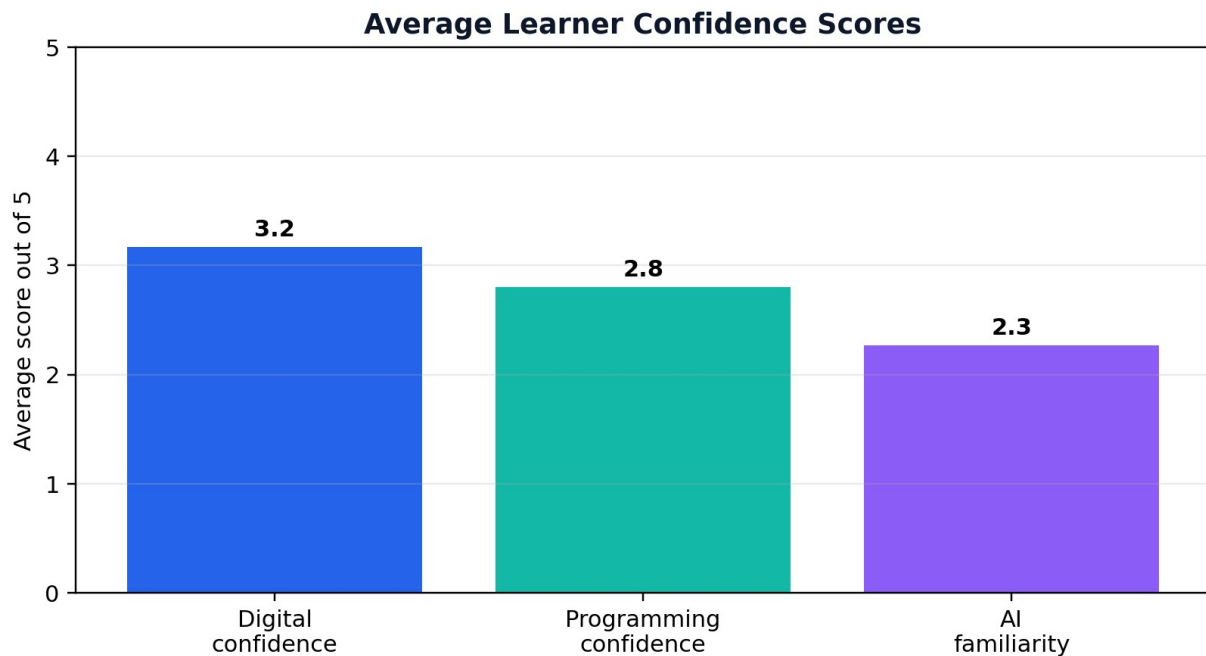


Figure 11: Confidence comparison across digital, programming and AI familiarity.

8. Prototype Description and UX/UI Design

The prototype is a full-stack application using a React + TypeScript frontend and an ASP.NET Core Web API backend. The user interface was designed as an academic intelligence dashboard: clean navigation, strong KPI hierarchy, clear risk badges, chart-first insight layout and a responsible-use layer before sensitive actions.

Solution Architecture

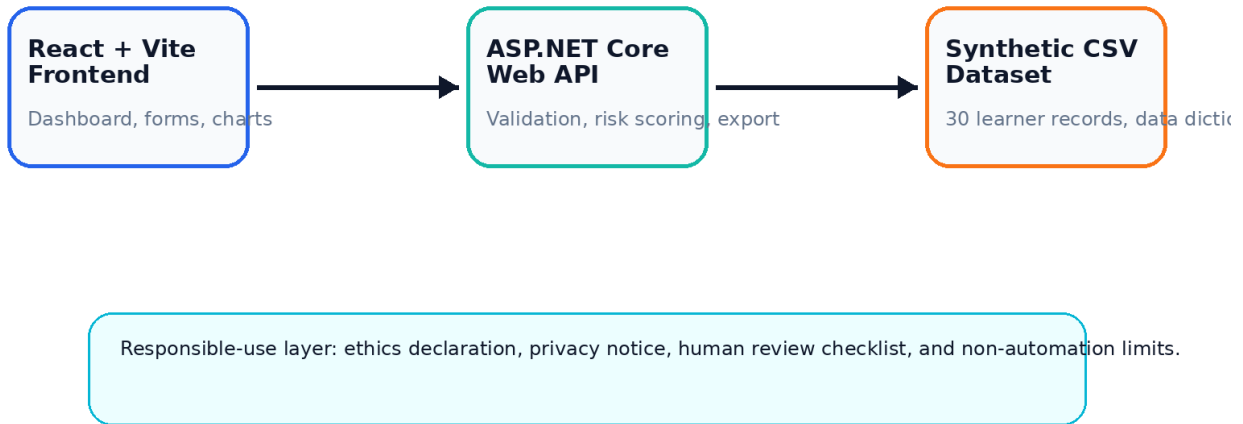


Figure 12: Full-stack solution architecture.

Feature	Implementation evidence	User value
CSV upload	POST /api/learners/upload and upload validation UI.	Fast import of learner survey records.
Manual learner entry	Form with inline validation.	Allows small corrections or direct capture.
Dashboard KPIs	DashboardSummaryDto and frontend KPI cards.	Programme-level decision support.
Five visualisations	Recharts and dashboard aggregates.	Pattern recognition for support planning.
Risk scoring	InsightsService weighted scoring.	Prioritised human review.
Recommendations	Recommendation API and cards.	Practical programme actions.
Ethics declaration	Frontend modal and ethics endpoint.	Reinforces responsible data use.
Export	CSV export endpoint.	Supports reporting and evidence sharing.

UX/UI Design Decisions

- High-level KPI cards appear first because programme staff need a quick operational summary.
- Risk categories use visual priority styling to support fast interpretation.
- Charts are grouped by learner risk, support demand, device access, internet access and confidence trends.
- The risk table exposes reason codes so users can understand why a learner was flagged.
- Ethics and privacy messages are placed inside the workflow rather than hidden in documentation.
- Help and settings views support usability during assessor demonstration.

Learner Risk Review

Risk scores are decision-support indicators and must be reviewed by staff before action is taken.

All



LEARNER ID	RISK LEVEL	KEY INDICATORS	SUPPORT NEEDED	ENGAGEMENT
02 L002	CRITICAL	Attendance High, Device Phone Only, Internet Unstable	Data;Transport	
04 L004	CRITICAL	Attendance High, Device Phone Only, Internet Limited	Data;Mentoring;Academic support	
07 L007	CRITICAL	Attendance High, Device Phone Only, Internet Limited	Data;Academic support	
09 L009	CRITICAL	Attendance High, Device Phone Only, Internet Unstable	Data;Transport;Mentoring	
12 L012	CRITICAL	Attendance High, Device Phone Only, Internet Limited	Data;Transport;Academic support	
14 L014	CRITICAL	Attendance High, Device Phone Only, Internet Limited	Data;Mentoring	
18 L018	CRITICAL	Attendance High, Device Phone Only, Internet Unstable	Data;Academic support	
20 L020	CRITICAL	Attendance High, Device Phone Only, Internet Limited	Data;Transport;Academic support	
23 L023	CRITICAL	Attendance High, Device Phone Only, Internet Unstable	Data;Mentoring	
25 L025	CRITICAL	Attendance High, Device Phone Only, Internet Limited	Data;Transport;Academic support;Mentoring	
29 L029	CRITICAL	Attendance High, Device Phone Only, Internet Limited	Data;Academic support	
32 L032	CRITICAL	Attendance High, Device Phone Only, Internet Unstable	Data;Transport;Mentoring	
34 L034	CRITICAL	Attendance High, Device Phone Only, Internet Limited	Data;Academic support;Mentoring	
38 L038	CRITICAL	Attendance High, Device Phone Only, Internet Limited	Data;Academic support;Transport	
40 L040	CRITICAL	Attendance High, Device Phone Only, Internet Unstable	Data;Transport;Academic support	

Figure 13: Learner risk review table with score, priority and reason codes.

Recommendations

Actionable interventions backed by current learner patterns.

Provide connectivity and device support

Rationale: 0 learners report limited or unstable internet and a meaningful share rely on phone-only or shared-device access.

Action: Prioritize data vouchers and device-lending slots for affected learners.

Launch targeted attendance support

Rationale: 0 learners match high-risk conditions (high attendance risk and confidence/access constraints).

Action: Assign weekly mentor check-ins and transport follow-up for flagged learners.

Introduce confidence-boost learning clinics

Rationale: 1 learners show low average confidence in digital/programming/AI skills.

Action: Run short practical labs and peer tutoring focused on foundational tasks.

Ethics And Privacy Panel

Responsible-use rules for staff reviewing support recommendations and learner risk outputs.

This tool is designed for synthetic or anonymized learner-support data. Do not upload direct personal identifiers, and limit data collection to fields required for support planning, review, and evidence reporting.

Learner data should be collected with informed consent, explained in plain language, and used only for agreed support-improvement purposes. Staff should be able to explain what data was captured, why it was captured, and how long it will be retained.

Risk scores and recommendations are decision-support signals, not automated decisions. Every intervention must be reviewed by a tutor, support officer, or programme lead who can validate context before action is taken.

Core Principles

- Data minimization: collect only fields needed for support planning and outcome review.
- Purpose limitation: do not reuse support data for unrelated disciplinary or admissions decisions.
- Fairness: interpret risk outputs as indicators of possible need, not evidence of learner failure.
- Transparency: keep scoring rules, recommendation logic, and data-quality limitations explainable to staff.
- Accountability: record who reviewed a recommendation, what evidence they considered, and what action was taken.

Human Review Checklist

- Verify data quality, missing values, and upload validation issues before acting on recommendations.
- Check attendance context, learner notes, and staff observations behind any high-risk flag.
- Confirm the proposed support action is equitable, proportionate, and aligned with learner consent.
- Record the rationale for interventions, human overrides, and follow-up outcomes for audit purposes.

Do Not Fully Automate

- Final intervention decisions for individual learners.
- Escalation, disciplinary, progression, or exclusion decisions.
- Interpretation of sensitive note content without human context.
- Any action that could materially affect learner opportunity without staff review.

Last updated: 2026-05-29

Figure 14: Recommendation and ethics evidence panel.

Manual Learner Entry

L

18-24

W

Laptop

Reliable

0

0

0

Employment status

Support need

Low

Add Learner

Figure 15: Manual form validation evidence.

9. AI, Data Ethics and Privacy Review

The prototype uses explainable, rule-based decision support. It is AI-enabled in the sense that it interprets support indicators and generates structured recommendations, but it does not make final decisions about learners. This distinction is important because learner support data can influence how staff perceive learners.

Data Sensitivity

The prototype avoids names, identity numbers, phone numbers, home addresses and unnecessary direct identifiers. However, support indicators such as attendance risk, internet access, device access and confidence scores are still sensitive because they can affect learner support decisions.

Responsible-Use Principles

Principle	Application in the prototype
Purpose limitation	Data is used only for learner support planning and programme improvement.
Data minimisation	The dataset avoids direct personal identifiers.
Transparency	Risk reason codes explain why a learner is flagged.
Fairness	Access barriers are treated as resource needs, not learner weakness.
Human review	Staff must review context before acting on flags or recommendations.
Harm prevention	The tool must not be used for exclusion, discipline, funding or progression decisions.
Retention control	Prototype data should be kept only for the assessment period or approved support cycle.
Security	Live deployment would require authentication, access control, audit logging and secure storage.

Human Review Workflow

6. Check whether the uploaded learner data is complete and valid.
7. Review the learner risk score, support need and reason codes.
8. Compare the dashboard signal with facilitator observations and learner context.
9. Choose a proportionate support action such as data support, mentoring or academic clinic referral.
10. Record the decision and follow-up date for accountability.
11. Review whether the support action improved learner participation or confidence.

Student Support Insights Tool

Responsible learner support analytics for programme teams

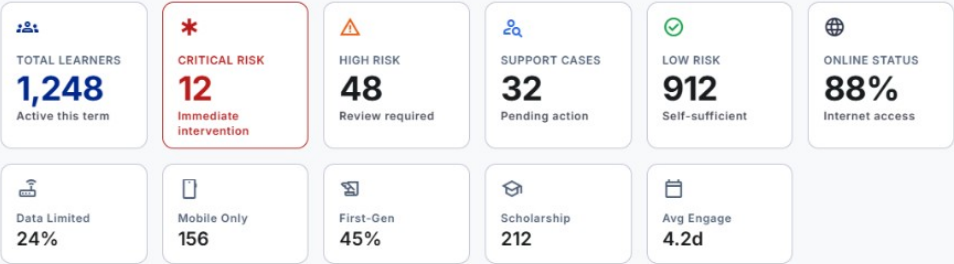
Ethics Declaration

Upload CSV

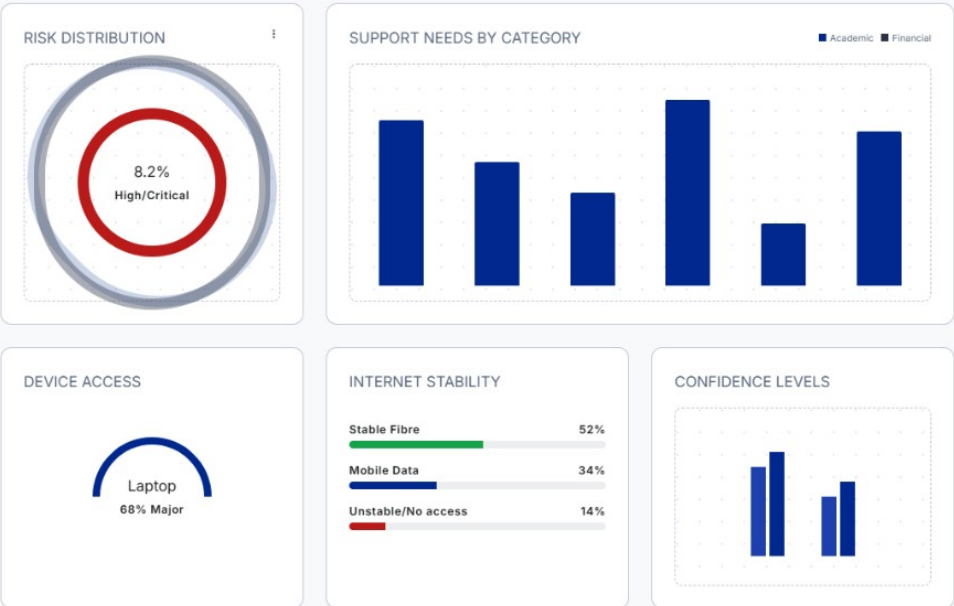
Add Learner

Export Report

Programme Support KPIs



Analytics and Visual Insights



Learner Risk Review

Search ID or Name

Filters

LEARNER NAME	RISK LEVEL	KEY INDICATORS	SUPPORT NEEDED	PROVINCE	ENGAGEMENT
SK S. Khumalo ST100452	CRITICAL	Low Connectivity, Zero LMS activity	Data Grant	KZN	
AM A. Mbeki ST100912	HIGH	Lacking hardware (Mobile only)	Academic Tutor	WC	
JN J. Naidoo ST100223	MEDIUM	Missing first assessment	Counseling	GP	
Showing 1-3 of 1,248 learners					
					Prev Next

Programme Recommendations

Implement Late-Night Data Support

PRIORITY: P1

Data Quality & Validation

OVERALL HEALTH

10. Methods, Tools and Development Approach

Development followed a staged delivery approach aligned with the assignment plan. The work moved from problem understanding and process modelling into data preparation, prototype build, dashboard development, risk logic, ethics review, testing and evidence packaging.

Area	Tool / method	Purpose
Frontend development	React, TypeScript, Vite	Build a responsive dashboard interface.
Visualisation	Recharts	Render dashboard charts.
Backend development	ASP.NET Core Web API	Provide validation, analytics, risk flags and export endpoints.
Data preparation	CSV dataset and data dictionary	Create safe synthetic learner data.
Version control / workflow	GitHub structure and CI workflow	Support quality gates and reproducible build evidence.
Testing	Manual tests, build tests and API checks	Verify prototype behaviour and evidence readiness.
Documentation	README, user guide, report, evidence folders	Make the submission traceable for assessment.

Development Workflow

12. Understand assignment requirements and define a realistic support analytics problem.
13. Create a synthetic learner dataset and data dictionary.
14. Model the current and improved learner support process.
15. Build backend validation, analytics and risk scoring endpoints.
16. Develop the frontend dashboard, charts, risk table and forms.
17. Add ethics declaration, privacy notice and responsible-use wording.
18. Test features, log bugs, improve quality and capture evidence screenshots.
19. Prepare the integrated report and map evidence to SFIA skills.

11. Testing, Review and Quality Evidence

Testing evidence covers endpoint availability, dashboard summary contract, transparent risk flags, CSV upload, validation errors, manual learner validation, ethics notice content, frontend type safety, frontend production build and backend release build. Bugs were logged and fixed with evidence of improvement.

Test area	Evidence	Result
Health endpoint	GET /api/health	Passed
Dashboard summary	GET /api/insights/dashboard	Passed
Risk flags	GET /api/insights/risk-flags	Passed
CSV upload happy path	Valid dataset upload	Passed
CSV upload validation	Invalid controlled values and malformed rows	Passed
Manual form validation	Missing or invalid values	Passed
Ethics notice	GET /api/ethics/notice	Passed
Frontend type safety	npm run typecheck	Passed

Frontend production build	npm run build	Passed
Backend release build	dotnet build configuration Release	Passed

```
Frontend Build Output

> student-support-insights-frontend@1.0.0 build
> vite build

@36mvite v5.4.21 @32mbuilding for production...@36m@39m
transforming...
@32mâ€”@39m 889 modules transformed.
rendering chunks...
computing gzip size...
@2mdist/@22m@32mindex.html @39m@1m@2m 0.59 kB@22m@1m@22m@2m â€”, gzip: 0.33 kB@22m
@2mdist/@22m@35massets/index-DQoLVBBw.css @39m@1m@2m 9.67 kB@22m@1m@22m@2m â€”, gzip: 2.93 kB@22m
@2mdist/@22m@36massets/DashboardCharts-D7mINDEj.js @39m@1m@2m 5.17 kB@22m@1m@22m@2m â€”, gzip: 1.76 kB@22m
@2mdist/@22m@36massets/LearnerForm-nzVAOahA.js @39m@1m@2m 5.52 kB@22m@1m@22m@2m â€”, gzip: 1.47 kB@22m
@2mdist/@22m@36massets/index-1sM-toP3.js @39m@1m@2m 19.58 kB@22m@1m@22m@2m â€”, gzip: 6.73 kB@22m
@2mdist/@22m@36massets/http-vendor-DcNlVx-A.js @39m@1m@2m 42.33 kB@22m@1m@22m@2m â€”, gzip: 16.66 kB@22m
@2mdist/@22m@36massets/charts-vendor-zIFtg3jN.js @39m@1m@2m264.25 kB@22m@1m@22m@2m â€”, gzip: 60.69 kB@22m
@2mdist/@22m@36massets/vendor-BojNP3JM.js @39m@1m@2m284.52 kB@22m@1m@22m@2m â€”, gzip: 94.72 kB@22m
@32mâ€” built in 3.83s@39m
```

Figure 17: Frontend build evidence.

```
Backend Release Build Output

Determining projects to restore...
Restored c:\Users\Yonela\New folder (3)\student-support-insights-tool\backend\StudentSupportInsights.csproj (in 1,86 sec).
StudentSupportInsights -> c:\Users\Yonela\New folder (3)\student-support-insights-tool\backend\bin\Release\net10.0\StudentSupportInsights.dll

Build succeeded.
    0 Warning(s)
    0 Error(s)

Time Elapsed 00:00:10.08
```

Figure 18: Backend build evidence.

Bug Fix Summary

Bug	Root cause	Fix applied
CSV notes containing commas failed import	Naive comma split broke quoted rows.	Quote-aware CSV parsing and column-count validation.
Risk output was difficult to justify	Risk flags had weak explanation metadata.	Weighted scoring, priority categories and reason codes.
Upload errors lacked field context	Errors were returned as flat strings.	Structured validation errors with line numbers and fields.
Dashboard contract was too limited	Older DTO supported fewer metrics.	Expanded dashboard DTO and richer KPI/charts.

Peer / Stakeholder Review

A review record is included to show that the value model, improved process, dashboard concept and risk/recommendation approach were reviewed. The key review outcome was to strengthen human-review wording and ensure the tool clearly presents risk scores as support signals rather than final decisions.

12. Hosting and Deployment Readiness

The project is prepared for cloud demonstration hosting. A split hosting model is recommended because the frontend is a static React application and the backend is an ASP.NET Core API. The project can be deployed using Azure Static Web Apps and Azure App Service, or using Netlify for the frontend and Railway for the backend.

Hosting option	Frontend	Backend	Best use
Azure option	Azure Static Web Apps	Azure App Service	Best for Microsoft cloud-aligned portfolio evidence.
Alternative option	Netlify	Railway	Fast demonstration hosting with simple GitHub-connected deployment.

Azure Hosting Plan

- Deploy the React/Vite frontend to Azure Static Web Apps.
- Deploy the ASP.NET Core backend to Azure App Service.
- Set VITE_API_BASE_URL to the hosted backend API URL.
- Restrict CORS to the hosted frontend origin.
- Use App Service settings for environment values and API keys.
- Capture hosted dashboard, health check, build and deployment screenshots as evidence.

Hosted Prototype Security Notice

The hosted version should be used with synthetic demonstration data only. If real learner data is ever used, the application must add authentication, role-based access control, protected learner/risk endpoints, audit logging, encrypted storage, retention rules and formal privacy governance.

Endpoint group	Demo status	Production requirement
Health and ethics endpoints	Can remain public.	Monitor availability and keep ethics notice accessible.
Learner, dashboard, risk and export endpoints	Acceptable for synthetic demo only.	Protect with authentication and role-based permissions.
Upload and manual entry endpoints	Should be gated by ethics declaration and API controls.	Require authenticated staff access and audit logging.

13. Reflection and Future Improvements

This project showed that a dashboard is only useful when the data, process and ethical controls around it are strong. The most important design decision was to make the risk model explainable. Instead of hiding the logic, the tool shows scores and reason codes so staff can review the context before taking action.

A key challenge was balancing prototype functionality with assessment evidence. The final submission therefore focuses on traceability: every major assignment requirement links to a section, file or screenshot in the evidence pack. This makes the project easier to assess and easier to improve.

Future Improvements

- Add secure user authentication and role-based access for real deployments.
- Add persistent storage with a controlled retention policy.
- Add intervention tracking so staff can record actions and outcomes.
- Calibrate the scoring model with anonymised historical programme data.
- Add accessibility testing and keyboard navigation evidence.
- Add automated backend unit tests and frontend component tests.
- Add a PDF summary export for management reporting.

14. Evidence Inventory Mapped to SFIA Skills

The evidence inventory shows how the project meets each SFIA capability area in the integrated assignment. This prevents the report from reading as separate tasks and gives the assessor a clear map between skill requirements and project evidence.

SFIA skill	Evidence in project	What it demonstrates
AIDE - AI and Data Ethics	Ethics review, governance checklist, ethics endpoint, privacy panel and human review wording.	Responsible data use, consent, fairness, harm prevention and non-automation boundaries.
BSMO - Business Modelling	Value model, stakeholder needs analysis and review evidence.	Business value, stakeholder alignment and decision-support outcomes.
BPRE - Business Process Improvement	As-is and to-be process models plus workflow explanation.	Process analysis, improvement opportunities and tool placement.
PROG - Programming / Software Development	Frontend, backend, validation, risk scoring, export and API endpoints.	Working prototype and implemented features.
DAAN - Data Analytics	Synthetic dataset, data dictionary, dashboard KPIs, charts, insights and recommendations.	Data preparation, analysis, visualisation and practical recommendations.
METL - Methods and Tools	README, user guide, workflow, CI, test log, bug log and evidence folder.	Structured development method, tool use and quality evidence.

15. Final Submission Checklist

Item	Status	Evidence note
Working prototype	Complete	React frontend + ASP.NET Core backend.
Integrated report	Complete	DOCX/PDF final report.
Synthetic dataset	Complete	30 learner records.
Data dictionary	Complete	Field definitions and values.
Three or more visualisations	Complete	Five visualisations included.
Five or more insights	Complete	Five key insights in analytics section.
Three or more recommendations	Complete	Connectivity/device, attendance support, clinics and resource review.
As-is and to-be process diagrams	Complete	Included and explained.
Business/value model	Complete	Value model and stakeholder mapping.
AI/data ethics review	Complete	Ethics review, privacy notice and governance checklist.
Testing and bug log	Complete	Test log and bug log included.

README/user guide	Complete	Run commands and user flow documented.
SFIA evidence mapping	Complete	Skill-to-evidence mapping included.
Screenshot evidence	Complete	Dashboard, validation, risk, ethics and build screenshots.

Appendices

Appendix A: Screenshot Evidence

EdusAnalytics

Support Insights Tool

Settings

Configure dashboard preferences, privacy-safe display options, and system behaviour.

Display Preferences

Theme Mode

Light

Table Density

Comfortable

Show Helper Text

Dashboard Preferences

Default Landing Section

Overview

KPI Visibility

Total Learners

Critical Risk

High Risk

Medium Risk

Low Risk

Internet Risk

Device Access Gap

Average Digital Confidence

Average Programming Confidence

AI Readiness Score

Data Quality Score

Top Support Need

Chart Visibility

Risk Distribution

Support Needs Breakdown

Device Access Distribution

Internet Access by Province

Confidence Comparison

Data & Validation Preferences

Validation Strictness

Standard

Highlight High-Risk Learners

Show Risk Reasons

Ethics & Privacy Preferences

Ethics Declaration Status: Confirmed

Show Responsible Use Reminder

Privacy Mode

Reset Ethics Declaration

Export Preferences

Export Format

CSV

Include Recommendations in Export

Appendix Figure A1: Settings view evidence.

EdusAnalytics

Learn Analytics

Dashboard

Learner Risk

Recommendations

Data Quality

Admin

Export

Student Support Insights Tool

Responsible learner support analytics for programme teams

Report Card

Add Learner

Export Report

Programme Support KPIs

10

11

4

15

50%

Help Centre

Guide to using the Student Support Insights dashboard responsibly.

How to use the dashboard

- Complete Ethics Declaration first to unlock upload, add learner, and export actions.
- Upload CSV data or add learner records manually, then review KPI and risk trends.
- Use filters in Learner Risk to prioritise human-reviewed interventions.
- Review recommendations and validation summary before export.

Ethics Declaration and privacy

- Ethics Declaration is mandatory and cannot be bypassed.
- Privacy Mode keeps output learner-safe and avoids unnecessary details.
- Risk outputs are decision support only and require staff judgement.

CSV upload requirements

Required fields include Learner ID, age band, province, device/internet access, confidence scores, employment status, support need, attendance risk, and notes. Upload accepts CSV only. Validation checks missing values, controlled value mismatches, and score ranges.

Risk score interpretation

Critical and High categories should be reviewed first. Reason codes show the contributing factors when enabled in Settings.

Export behaviour

Export stays disabled until Ethics Declaration is confirmed and learner data is loaded. Settings control whether recommendations, validation summary, and ethics status are included.

Troubleshooting API connection

- Check backend is running on `http://localhost:5000`.
- Use Settings - Check API Status to confirm readiness.
- If API is offline, dashboard cards and charts show empty-state placeholders.

Contact / support

Support channel placeholder: `programme-support@example.edu`

Close

1000	10000	Attendance High: Device Phone Only, Internet	Yes	--	Download
1000	10000	Attendance High: Device Phone Only, Internet	Yes	--	Download
1000	10000	Attendance High: Device Phone Only, Internet	Yes	--	Download

Page 28

Appendix B: Risk Review Extract

Learner ID	Province	Attendance	Support need	Score	Priority
L002	Gauteng	High	Data	100	Critical
L006	KZN	High	Data	100	Critical
L009	KZN	High	Data	100	Critical
L013	Western Cape	High	Data	100	Critical
L017	Gauteng	High	Data	100	Critical
L021	KZN	High	Data	100	Critical
L025	Western Cape	High	Data	100	Critical
L029	Gauteng	High	Data	100	Critical
L005	Gauteng	High	Transport	98	Critical
L015	KZN	Medium	Academic support	75	High

Appendix C: Improvement Backlog for Future Development

Improvement	Reason	Priority
Secure authentication and role-based access	Required before using real learner data.	High
Protect learner and risk read endpoints	Prevents public exposure of support indicators.	High
Persist intervention follow-up actions	Allows staff to track support outcomes.	Medium
Calibrate risk scoring with approved anonymised data	Improves reliability beyond synthetic data.	Medium
Automated regression tests	Strengthens engineering evidence.	Medium
PDF export of management summary	Improves reporting usefulness.	Low